

Preemptive Cybersecurity for the Age of AI-Enabled Fast Moving Attacks

Mathematically Rigorous — Preemptive, Complete, Continuous,
and Verifiable — Cyberdefense Management



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1. INTRODUCTION: The Modern Threat Landscape is Catastrophic

1.1. AI-Based Fast Moving Attacks Pose Existential Risks to Enterprises

Imagine a scenario where fast-moving AI powered cyber attacks breach your state-of-the-art cyber defenses and cause catastrophic damage. By the time an attack is detected, it is already too late. This is not hypothetical; in September 2025, a hack by Chinese state-sponsored hackers jailbroke Claude Code, Anthropic's agentic AI. Using Claude Code, they executed a sophisticated espionage campaign against roughly 30 targets, operating mostly autonomously and executing the attack at speeds that humans cannot physically match¹. Thus there is an urgent need for truly preemptive cyberdefense to protect organizations well *before* a breach can occur.

The consequences of failing to rise to this challenge are catastrophic: In 2025 alone, cybercrime was projected to cost up to 9.5% of world GDP (\$10.5 trillion globally)², cyberattacks surged over 200%³, breach times collapsed to under an hour⁴ and, most critically, >60% of breaches involved human factors⁵.

1.2. These Challenges Have Only Intensified in the Post Mythos Era

With the release and demonstration of AI models with unprecedented autonomous vulnerability detection and exploitation capabilities such as Anthropic's Mythos, cyber defense has entered a fundamentally new era. Threats are evolving faster than vulnerability disclosures and patch cycles can respond. Attackers are now exploiting

¹ <https://www.anthropic.com/news/disrupting-AI-espionage>

² <https://cybersecurityventures.com/cybercrime-damage-costs-10-trillion-by-2025/>

³ https://drj.com/industry_news/slashnexts-2024-phishing-intelligence-report-shows-credential-phishing-attacks-increased-by-703-in-the-second-half-of-the-year/2019-2024

⁴ <https://ir.crowdstrike.com/news-releases/news-release-details/2024-crowdstrike-global-threat-report-breakout-breach-under>

⁵ <https://www.verizon.com/business/resources/reports/dbir/>



vulnerabilities prior to public disclosure and chaining together low-severity weaknesses to breach environments that appear protected against critical threats.

1.3. Frontline Defensive Controls Are Even More Critical for Defense

In such an environment, patching critical CVEs may no longer be practical. In many cases, no CVE has been published or, if it has, a patch may not yet be available, due to the sheer volume of reported vulnerabilities. Frontline cyberdefense controls, paired with the most up to date threat intelligence, are the primary solution to blocking such new known attacks.

Therefore, enterprises must continuously ensure the correctness of the tens of thousands of static configurations and dynamic state in their frontline security controls, including network segmentation, security hygiene, best practices, access boundaries, and defensive policies, keeping pace with the latest available threat intelligence.

2. PROBLEM: Traditional Approaches Are Incapable of Meeting the Challenge

The cybersecurity industry has invested heavily in data path cyber defense vendor devices such as next-generation firewalls (NGFWs), endpoint protection platform (EPP), zero trust network access (ZTNA), web application firewalls (WAFs) etc. The problem lies not in their capability, but in ensuring that defensive devices have the correct posture in both static configurations and dynamic state.

Gartner estimates that **99% of firewall breaches are caused by misconfiguration**, including weak or missing configurations, not flaws in the firewall itself⁶. Gartner further predicts that through 2029, more than **60% of all security incidents will be traced to misconfigured** technical security controls⁷.

⁶ <https://www.gartner.com/smarterwithgartner/is-the-cloud-secure>

⁷ <https://www.gartner.com/en/documents/5718951>



Indeed, our analysis confirms this (see [Appendix 1](#)). Simply having state of the art cyber defense tools is insufficient; had a breach occurred, it would have been traced back to one of these misconfigurations.

Several recent high-profile breaches also seem to bear out this statistic. The 2023 MOVEit attack compromised over 2,700 organizations and exposed data on 93.3 million individuals⁸, many of which were compromised weeks or months after the vulnerability was published. By January 2024, the "Mother of All Breaches" exposed 26 billion records due to a firewall misconfiguration⁹. The Knownsec leak in late 2025 revealed that even state-sponsored attackers rely heavily on exploiting configuration gaps and existing vulnerabilities rather than developing novel exploits¹⁰.

2.1. There Are an Astronomically Large Number of Possible Cyber Attacks

The challenge is that verifying the cyberdefense posture is intractable with existing approaches. Modern cyberdefense devices must examine network, protocol, session, and application headers as well as their content. If we assume that they examine ~1000 bytes (8000 bits) per session to ensure that it is truly protecting organizations and users from malicious traffic, we arrive at an astronomically large 2^{8000} state space.

⁸ <https://www.emsisoft.com/en/blog/44123/unpacking-the-moveit-breach-statistics-and-analysis/>

⁹ <https://cybernews.com/security/billions-passwords-credentials-leaked-mother-of-all-breaches/>

¹⁰ <https://cybernews.com/security/knownsec-leak-exposes-involvement-in-state-linked-cyber-operations/>

2.2. Current Approaches Are Either Reactive or Incomplete

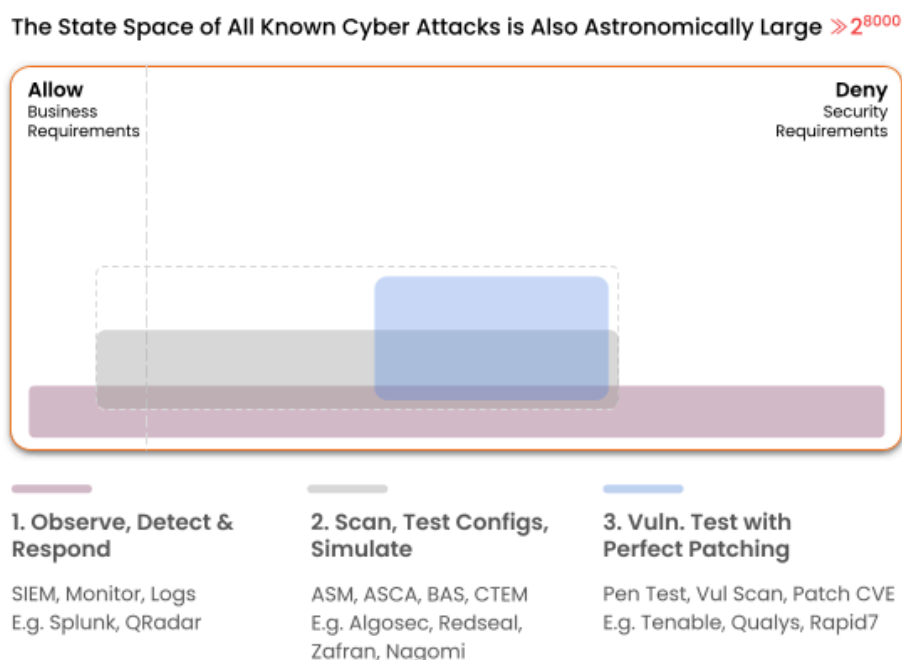


Figure 1: State Space of Cyber Attacks (not to scale): Cyber devices might examine ~1000 bytes (8000 bits) of network, protocol, session, and application headers as well as content to maintain adequate cyberdefense. Monitoring (pink box) is critical, but is reactive and only captures a small amount of information. Scanning methods (grey box) can cover a larger portion of the state space but are resource-intensive. The dotted grey box shows that larger, better-resourced organizations may expand their scan coverage but are ultimately still resource-limited. Vulnerability testing with perfect patching (blue box) is not only untenable, but even so, can only effectively neutralize the ~20% of attacks that are vulnerability based, but cannot address the remaining 80% of known attacks that do not exploit vulnerabilities. Rigor can address a much larger fraction of the cyber defense space at the configuration level by mathematically modeling cyber defenses, ensuring that all findings about policies and defensive configurations are deterministic and verifiable.

Traditional approaches cannot cover this entire space with any measure of resource efficiency (see Figure 1).



1. *Observation and Detect & Respond tools (SIEM, monitoring, logs)*: These are essential but fundamentally reactive, addressing problems only after an attack occurs.
2. *Scan, Test, and Simulate tools (ASM, ASCA, breach and attack simulation)*: These preemptively test configurations against known attack patterns but can only sample a very small fraction of the state space before they impose an untenable resource load on the enterprise.
3. *Vulnerability Management (with pen testing, CVE patching)*: While these preemptively address known vulnerabilities, achieving perfect and continuous patching, even for only critical vulnerabilities, is often impractical. Besides, known exploited vulnerabilities (KEVs) are only exploited in ~20% of all known attacks, leaving non-vulnerability attack vectors unaddressed.

Even in combination, these approaches leave significant gaps, without providing guarantees about protection against known attacks.

2.3. Addressing This Challenge Requires a Radically Different Solution

Given the astronomically large $\gg 2^{8000}$ state space of potential attack traffic, traditional reactive and sampling-based security approaches, even when combined, leave critical structural gaps without providing verifiable protection against all known threats. These unverified structural weaknesses are precisely what modern, AI-enabled attackers exploit to achieve rapid, catastrophic breaches. For instance, the PUNK SPIDER campaign bypassed endpoint security by compromising an unmanaged IoT webcam and leveraging overly permissive cross-VLAN routing to laterally access and encrypt corporate SMB file servers. Because existing tools cannot *prove* network segmentation is correct, this type of structural misconfiguration is exploited before any alarm can be raised.

To meet this existential challenge, we must move beyond heuristics and sampling. Addressing this requires a mathematically rigorous solution that ensures cyber defenses are configured correctly against all known attacks. This is the **only way** to gain certainty against the complexity of the challenge of the astronomically large 2^{8000} state space. Defense posture must be **preemptive**, identifying and remediating control gaps



before attackers can exploit them, **complete**, ensuring protection against all or at least as many known attacks as possible, and **continuous**, securing defense postures to match the latest threat intelligence data as it becomes available. Finally, enterprises must trust nobody, investing instead in solutions that are independently **verifiable**, underpinned by mathematical proofs, to give them certainty about their defensive postures.

3. INNOVATION: We Are Pioneering A Rigorous Cyber Defense Management Platform

Building a mathematically rigorous cyberdefense platform requires solving three extremely hard problems **in tandem** to span the space of attacks and defenses.

3.1. Attack: MITRE-Enriched Models of All Known Cyber Attacks

Threat intelligence exists in terabytes of semi-structured and unstructured data—advisories, research reports, indicators of compromise (IOCs), indicators of attack (IOAs), and other sources used by SOCs and product security incident response teams (PSIRTs). We must ingest threat intelligence from a variety of sources to capture deep dependencies and produce enriched, fine-grained models of known attacks following the MITRE ATT&CK framework. These models must capture the Tactics, Techniques, and Procedures (TTPs) of known threat campaigns against which defense models can be compared precisely to determine whether an organization is truly protected.

Rigor's solution brings together cross-domain capabilities to solve this problem, combining well-understood machine learning (ML) and natural language processing (NLP) techniques with modern developments in large language models (LLMs), which have only recently become cost-efficient and feasible for large-scale use, to produce deep, detailed, granular attack graphs for every threat that matters to the customer.

3.2. Defend: Symbolic Model of Computation of Cyber Defenses

Modern cyber devices have 100K+ configurations and millions of dynamic states. Identifying control gaps with certainty requires exhaustive formal mathematical models



of these devices — models that resolve static objects, dynamic operational state, IPs and ports, external objects and lists, users and groups, fully qualified domain names (FQDNs), regular expressions, temporal fields, and more. Building formal models at scale, across vendors, is an engineering challenge that *no vendor has achieved before!* It requires large scale compute, hundreds of gigabytes of memory, and demands deep domain knowledge in cybersecurity and expertise in formal methods and AI-based automated reasoners.

Rigor's Defense Intelligence pillar builds on twenty years of experience building and delivering formal methods to over 4000 marquee customers to protect against adversarial router attacks, ensuring network memory performance, assure network and, data-center security and access security at scale, combining exhaustive formal behavioral models of customer configurations with a powerful universal query language that allows for pinpoint determination of control gaps in defensive configurations against a given attack.

3.3. Remediate: Guardrailed Reasoner Built with Grounded World Models

The system must then query the defensive model for control gaps against every attack that is modeled. Once a control gap has been identified, we must use this knowledge to prescribe the right configurations without introducing new errors, and to do so repeatably and reliably for every defensive control against every attack.

For this final piece, **Rigor AI delivers a semantic reasoner**, augmented with AI agents that are trained in the detailed ontology of cyberdefense, and on each cyberdefense datapath vendor's behavior and capabilities. The AI reasoners are aware of both the attack and defense models, as well as the organization's business, regulatory, and infosec requirements, and prescribes precise, vendor-specific, context-aware, and risk-prioritized remediations to ensure that configurations are updated to maximize defensive protection while maintaining service availability. Reasoners are equipped with a powerful custom query language that enables them to pinpoint adherence to properties and constraints. Precision in remediation is achieved using several techniques such as guardrailing, chain of thought reasoning, reflection, and the use of

formal methods to build world models to ground the reasoners and prevent hallucinations.

Figure 2 shows how these three intelligence pillars make up a fully preemptive cyberdefense system. The system takes customer defenses and configurations as inputs, alongside intelligence about a curated set of attacks that matter to the customer. Once control gaps are identified and verified remediation recommendations are generated, the network security or SOC admin may review these recommendations and determine which ones to apply, or even choose to run them *autonomously*.

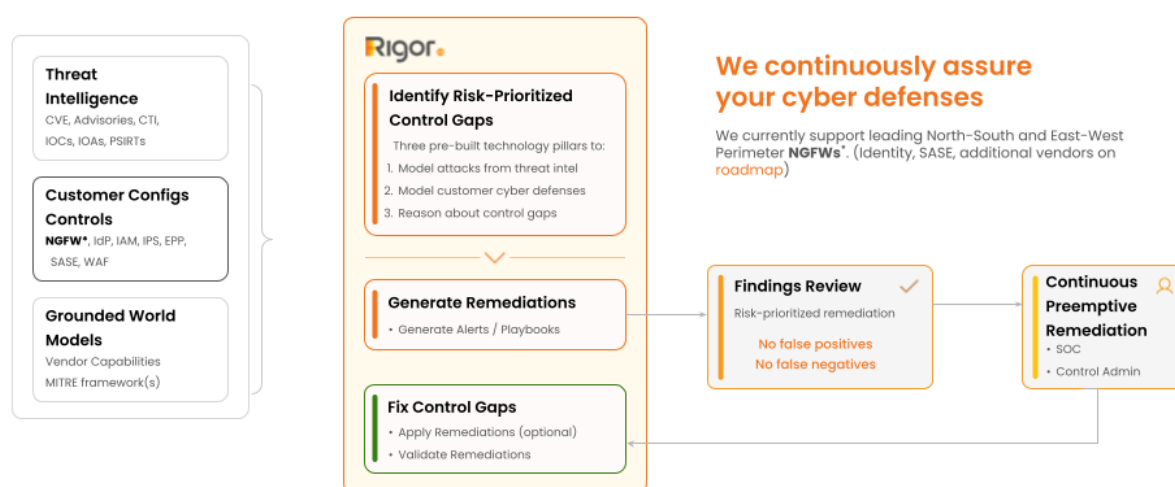


Figure 2: RigorOS Management Plane Platform Operational Flow: The RigorOS platform collects threat intelligence inputs alongside technology controls configurations and vendor capability information. The three pre-built technology pillars ingest this data to produce deeply grounded world models. These world models are learned representations of attack patterns, defense mechanisms, and system behaviors that make up the operation of the cybersecurity domain. These models are anchored in structured domain knowledge, e.g. MITRE ATT&CK and D3FEND ontologies and TCP/IP protocols, and real-world artifacts, e.g. vendor datasheets to provide deep technical and mathematical grounding. Our foundational use case, described below, assures firewall configurations, with additional use cases extending this to complete assurance of other cybersecurity domains.

4. SOLUTION: Rigorous Cyber Defense Delivered on Formally Verified NGFW Assessments

Rigor is launching the world's first preemptive cyberdefense management platform by prioritizing Next-Generation Firewall (NGFW) policy assessments, as they constitute the first and most critical line of defense (see Figure 3). NGFWs are sophisticated devices that inspect L2-L7 network traffic, and their many controls, such as network segmentation, access boundaries, and malware detection, are capable of preventing nearly 80% of cyber attacks¹¹. Moreover they are essential for containing the blast radius in the event of a breach.

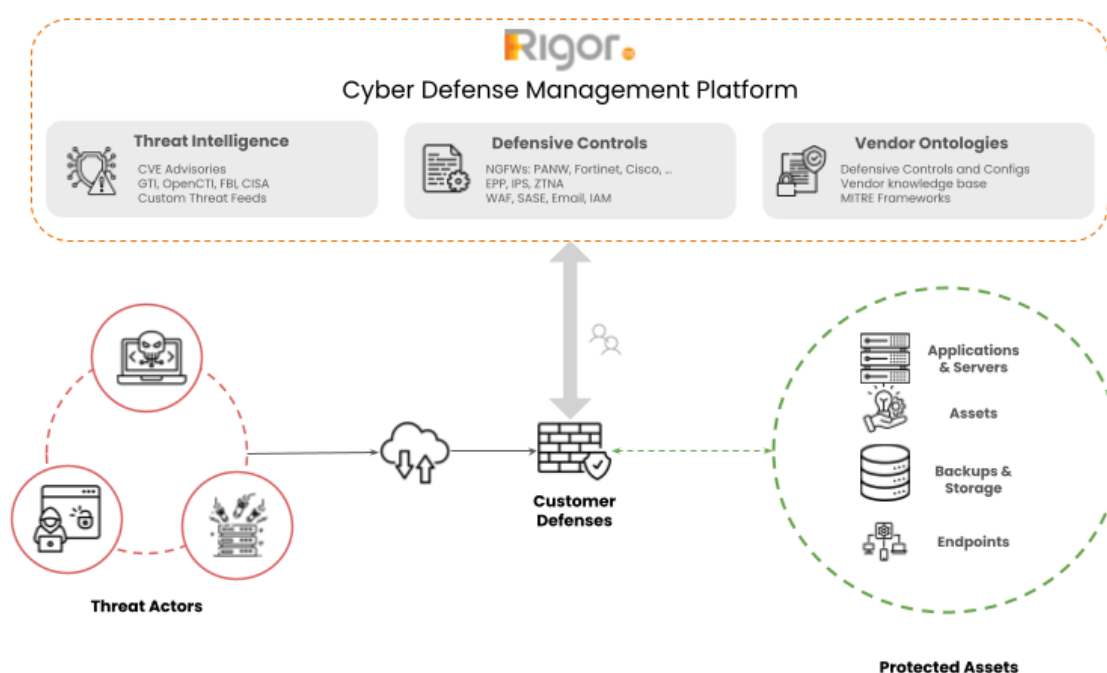


Figure 3: RigorOS Management Plane Platform Topology Flow: Rigorous cyber defense management of NGFWs. EPP, ZTNA, WAF, IAM on our roadmap. We offer assessments through a SaaS platform, with limited support for on-prem and airgapped environments.

¹¹ <https://www.webspy.com/blog/80-of-cyber-attacks-are-preventable/>



4.1. Comprehensively Modeling NGFWs, Leaving No Stone Unturned

The magic begins when we build a formal mathematical model of the customer's cyber defense. Rigor's firewall assessment service models firewall configurations comprehensively and exhaustively and identifies security gaps for posture hardening, conflicting rules, and optimization opportunities through redundancy elimination and rule consolidation.

We comprehensively model and analyze configurations, L2-L7 security policy, dynamic objects including fully qualified domain names (FQDNs), profiles, actions, user and group configurations, policy schedules, predefined DBs, geolocation data, policy statistics, and intent.

By building the deepest and most comprehensive models in the industry, we can deterministically pinpoint several categories of syntactic, typo, semantic, behavioral, and topological rule errors, weak rules, intent errors, translation errors, redundant rules, shadows, pair-wise conflicts, multi-rule conflicts, multi-device errors, synchronization errors and much more. This mathematical model provides the basis for every subsequent query and analysis.

4.2. Incorporating Best-In-Class Threat Intel for Continuous, Targeted Cyber Defense

We build deep MITRE enriched cyber attack models from real-time access to petabytes of the most up to date known threat intelligence. This includes open source, CTI, and paid threat intelligence such as Google Threat Intelligence. We then query the formal model of a firewall (and eventually of multiple cyber defense controls) to ensure it is configured to protect against different TTPs for any given attack.

Combining formal models, which give us a detailed and complete view of the customer's cyberdefense configurations, with threat models enables us to generate risk-prioritized remediations for every control gap we identify, against every threat that matters.



4.3. Maximizing Your Defensive Controls By Using Residual NGFW Capacity to Block Attacks

The platform is designed to programmatically ingest and understand threat intelligence and immediately provide actionable policy recommendations to protect against threats. This allows us to use the residual capacity of our customers' existing cyberdefenses to add threat blocking rules and further fortify their defenses. By doing so, we help customers extract the maximum value from their existing tools.

Contrast this with the more common and traditional reactive approach in which enterprises often don't update rules based on threat intelligence until they've already detected suspicious activity. With the increasing frequency of attacks and the rapidity of escalation, this is likely too late to be useful in defense.

Where threat intelligence has traditionally lived in the Detect and Response pillar with SOC and threat hunting teams, we bring its value to preemptive cyber defense and the front line of security in the post-Mythos age.

4.4. Pioneering a Powerful Symbolic Formal Query Language

We build a powerful symbolic language by using and augmenting SAT, SMT solvers. These are used to build structured models specific to each customer's NGFW. Rigor further integrates NLP and LLMs to augment the symbolic language. These can support powerful queries that can be used to check for adherence to various properties and constraints. These queries include but are not limited to network segmentation, reachability, business and infosec mandates, regulatory compliance. We then ensure that the customer's firewall posture formally adheres to these compliance constraints, precisely, and at all times, which leads to a variety of use cases as shown in the next section.

4.5. Bringing Deep Actionable Value to Customers In <15 Days

We build these capabilities into every assessment, and provide our customers with detailed remediation recommendations in under 15 days, and for less than 15 minutes of their time. These recommendations are immediately actionable, helping customers fortify their cyber defense posture preemptively and verifiably.



5. USE CASES: The RigorOS Platform Delivers Several Use Cases

As described in the previous section, Rigor *builds a formal mathematical model* of the customer's cyber defense. Depending on the stimulus being sent to the mathematical model, Rigor delivers a suite of preemptive cyberdefense use cases on our RigorOS platform (see Table 1). This section describes some groundbreaking benefits that RigorOS delivers to customers, while the next section covers a broad roadmap.

#	Risk	Customer Problem	Use Case
1	Security Posture	Are my critical segmentation, security inspection configurations correct?	Formally Verified Policy Analysis
2	External Attack Surface	Are my North-South and East-West perimeter surface controls operating as per my operational & security requirements?	Formally Verified ASM Compliance
3	Change Mgmt.	Are my post-change security and business posture meeting the change intent?	Formally Verified Change Management
4	Config Drift Mgmt.	Have my configurations drifted away from my approved golden baseline?	Configuration Drift Management
5	Vendor, Solution Migration	Are my pre- and post-migration security and business posture equivalent?	Formally Verified Vendor Migration
6	Business, Regulatory Infosec	Are my defensive controls operating according to my business, regulatory and infosec requirements?	Formally Verified Compliance
7	Cyber Attacks	Are my defensive controls protecting me from known attacks that matter to me?	Preemptive and Complete Cyber Security
8	Partner Supply Chain	Are my partners defensive controls protecting me against known attacks that matter to me?	Preemptive 3rd-party Risk Analysis

Table 1: RigorOS Use Cases: The RigorOS platform is a game changer bringing an unparalleled and rigorous methodology to several posture, operational, compliance, and security use cases. Use Cases 2-8 all build on the foundational use case, Formal Security Posture Verification.



6. VALUE: Rigor's Assessments Have Already Benefited Several Marquee Customers

The Rigor assessment service is already operational with and has been deeply impactful for design partners and paying customers across industries and sectors: financial services, critical infrastructure, state and local government, technology, and education. By enriching our policy findings with key threat, compliance, and other information, we provide

Below, we present a few example findings from recent customer firewall analyses, from formal verifications of firewall configurations to surface lurking errors that human review and existing tools had not caught, to verifying attack surface scans, to helping our customers ensure compliance with local and federal mandates.

6.1. Preemptively Remediated Incorrect Posture That Allowed GitHub Credential and IP Theft

A deep lurking human error in app-control inspection profiles for the unsanctioned app GitHub inadvertently allowed access to the app from some subnets, violating the CISO intent.

Specifically, this FinTech customer has a security requirement to block an unsanctioned app, GitHub. A high priority rule allows access to GitHub from some internal objects which are resolved by Rigor's formal analysis into specific source IPs, and a second slightly lower priority rule also allows access to a GitHub repository from a different set of internal source IPs resolved by Rigor's analysis. A much lower rule blocks access from internal IPs to specific blocked domains, including GitHub, per the CISO's intent for the organization.

Further, the address object blocks only `github.com*`, and does not include subdomains (`*.github.com*`) that may be used to access, e.g. GitHub APIs. Thus, not only is the rule blocking the domain partially shadowed, it is also potentially incomplete.

Due to the use of incorrect objects and the priority ordering, the security requirement to block github from internal source IPs was being violated. Risks to the company include



intellectual property loss and credential exposure, which our detailed findings helped the customer address and mitigate.

6.2. Preemptively Closed SMB Lateral Migration Attack Paths

SMB app-control profiles were completely shadowed by a hard to find combination of several interspersed rules, allowing vulnerable SMBv1.0 file transfer to pass unchecked, exposing the customer to lateral malware migration.

In this case, a low priority on the customer's NGFW is configured for disaster recovery file access using the SMB protocol, including IPS and AntiVirus, as intended. However, a combination of three higher priority rules formed a very hard-to-uncover group shadow allowing traffic from different company branches to HQ with weak security profiles, conflicting with the intent. This violates security requirements on IPS and AntiVirus on SMB traffic, and resulted in improper protection against SMB v1.0 traffic, resulting in an open and unprotected lateral movement path for malware.

6.3. Preemptively Assured EASM Compliance for European Government Agency

External Attack Surface Management (EASM) is a critical use case for large enterprises and nation states but is fundamentally incomplete; even something as simple as source IPs cannot be simulated on the Internet! Rigor symbolically queries the formal models of the customer's defense posture to augment the EASM scan and provide a **complete and detailed** picture of the organization's external attack surface, in a way that even state-of-the-art EASM cannot provide.

A European country's cyber defense umbrella agency was performing external attack surface scans (EASM) on their government agencies. Rigor was able to perform EASM scans, and show how the coverage of these EASM scans could be formally extended beyond the capability of traditional EASM, and also formally check if the external attack surface of these agencies were formally compliant.



6.4. Preemptively Identified Regulatory Compliance Failures From Malicious IOC Exposure

A regulated Industry customer in Asia had a compliance requirement to block malicious IPs and domains from the nation state's cybersecurity agency, received daily via email. Rigor was able to model this email based threat feed using our LLMs developed for attack analysis. We performed an additional formal analysis of the customers posture, with more than 3.5K rule match and security action combinations, and 50 Million address combinations. We were able to preemptively prove that there was *no malicious IP protection at all* for traffic going to some internal IPs which put the customer at risk of regulatory fines and exposure to attacks and breaches.

6.5. Preemptively Diagnosed Infosec Compliance Failures

Rigor performed a compliance assessment for a North American State University with thousands of rules. Amongst several findings, Rigor's formal models demonstrated that certain endpoints in their trusted zones, were exposed to Internet facing traffic without the proper anti-virus, anti-vulnerability profiles, and virus filtering on their NGFW firewalls.

Our deep formal models provide a level of granular visibility into every defensive configuration that no other approach in industry can match. Through this, the platform automatically provides extremely detailed and specific types of findings to provide exhaustive coverage of all the customers' security posture. These findings can be used for policy reduction, hardening, and conflict resolution.

Additional customer Findings snippets are found in Appendix 1.



7. ROADMAP: Use Cases Under Development

The following use cases are under active development with limited availability through our assessment service.

7.1. Configuration Drift Management

Formal modeling allows security administrators to save a “golden baseline” state against which they can compare future firewall states. Even the most microscopic change in the configuration, e.g. a single port on a firewall being switched from allowing to denying traffic, or vice versa, shows up as a difference in a model, allowing admins to manage drift in their security configurations at any time, and from any configuration change.

7.2. Formally Verified Change Management

We also extend our formal verification to ensure that a change in the security posture meets the intent of the change completely by modeling the new intent as a requirement that the formal behavioral model is queried against.

7.3. Formally Verified Vendor Migration

When customers migrate from one firewall vendor to another, formal analysis ensures that the security posture after migration is equivalent to the posture before, accounting for differences in the way different vendors handle each security function.

7.4. Preemptive 3rd Party Supply Chain Risk Analysis

We can further extend our preemptive cybersecurity platform to model the defenses of an organization’s 3rd party supply chain, ensuring that this attack vector for an organization is neutralized.



8. EXPERTISE: We Leverage Decades of Industry-Leading Experience

The industry is already anticipating a move toward widespread deployment of preemptive cybersecurity solutions. Gartner projects that by 2030, preemptive cybersecurity solutions will account for 50% of IT security spending, up from less than 5% in 2024¹². The economics of such a shift are straightforward: blocking an attack before it enters costs a fraction of breach response, remediation, and recovery.

Rigor is leading the charge toward a truly rigorous cyber defense future, leaning heavily on the team of experts we have assembled to bring this vision to fruition. Our founding team includes Silicon Valley entrepreneurs, academics, ex-military intelligence professionals, distinguished alumni from Stanford, CMU, UT Austin, and the IITs and veterans from Google, Cisco, Microsoft, Meta, Palo Alto Networks, VMware, Intel, Symantec, F5, Service Now, Lockheed Martin, and other well-known companies. We collectively represent over 700 years of experience in engineering, research, and enterprise product development. We have previously secured data center, enterprise, and access networks for over 4000 customers including Fortune 500 companies and government security agencies, representing over 10 successful startups, and over \$1.5 billion in past revenue.

Having spent the last decade bringing formal methods-based network security to data centers, we turn our sights on preemptive cyber security. With several use cases already delivered to early customers and more in active development, we are poised to deliver enormous value to customers in this new space.

9. JOIN US: On a Rigorous Cyber Defense Journey!

Rigor has already engaged with 20+ customers and design partners across federal and state government, critical infrastructure, regulated industry, and MSSPs. These include F500s across sectors, trillion-dollar companies in AI and hyperscale computing, category-leading entities in energy, digital health, pharmaceuticals, telecom and banking, a leading state university, critical mobile communications providers, and

¹² [Gartner Press Release, September 18, 2025](#)



nation states. Engagements span North America, Europe, Middle East, South Asia, ASEAN, and Pacific.

We are expanding our offerings to additional early customers through design partnerships, pilot deployments, and full production contracts. We offer our assessments through a SaaS platform, with on-premise and air-gapped deployments in development for customers with varied data isolation and sovereignty needs. By integrating with ticketing systems (e.g. ServiceNow, JIRA), we maintain complete explainability of our AI, traceability of formal proofs, and system logs so that customers may verify our findings themselves at every step rather than trusting us blindly.

We demonstrate the value of RigorOS with a < 15 minute time commitment from customers, and in less than 15 days. To provide a foundational security posture verification, we build a mathematical model of computation of our customers' cyber defenses on which we deliver several additional use cases; the initial assessment is just the first step in a cyber defense journey, an investment in your security and long-term savings. Ongoing developments in the platform translate to more options for the customer: first on-demand assessments, then continuous assessments, more use cases, additional device coverage, and other expanding capabilities, including reporting and audits. To kick off your journey with your first formal policy analysis, contact us at info@rigor.ai.



Appendix 1: Example Findings and Impacts from Formal Analysis of Customer Configurations

Mitigated criminal liability for CISO

Fintech | Asia

Misconfigurations caused rules permitting internal hosts to connect without IPS, antivirus, web filtering, DNS filtering, or app control, bypassing the CISO intent to block a list of malicious hosts received from the government's cybersecurity threat intelligence agency.

Mitigated several critical NIST, ISO non compliance issues

Critical Infrastructure | North America

Licensing, static configurations including decryption and header decryption, and dynamic antivirus signature updates were correctly enabled on but a misconfiguration in deep SSL decryption rendered IPS and virus filtering ineffective and resulted in the mandate for IPS and antivirus not being met.

Mitigated disruption of business-critical SLAs

Telecommunications | Southeast Asia

HTTPS traffic rules for critical hosts were completely shadowed by broader geo-blocking rules. With infrequent traffic patterns, this went undetected. Rigor identified these unreachable rules, preventing business-critical traffic disruption.

Hardened overall security posture for trusted networks

State, Local, and Education (SLED) | North America

The customer had ~25% of security rules for internet facing traffic configured with inadequate, weak, or no threat inspection profiles. Formal analysis revealed these rules as reachable, potentially allowing malicious traffic into trusted networks.

Hardened posture to prevent malware, phishing, exfiltration

Bank | East Asia

Inconsistent threat inspections across collaboration apps, allowed Teams with only to URL filtering while WebEx had comprehensive threat inspections. We revealed that



Teams users were exposed to malware, phishing, and data exfiltration risks.

Eliminated exposure to known CVEs for critical servers

Critical Communication Infrastructure | North America

IPS filters were missing on vulnerable rsync protocol traffic to a few key backup servers, putting the enterprise at risk for lateral movement of malware and exposure to several known CVEs.

Closed remote code execution attack paths

Regional Bank | East Asia

Both PANW and Fortinet NGFWs allowed Windows server update service traffic from trusted zones with no IPS inspection, despite both vendors providing protection from vulnerability signatures. This exposed the bank to remote code execution attacks on critical infrastructure across their multi-path, multi-vendor environment.

Compressed rules by 45% making NGFW easier to manage

Education | North America

The NGFW configuration included over 1,800 hard-to-identify redundant and mergeable rules with complex hierarchical dynamic objects and profiles, interspersed in a ~4000 rule configuration of their NGFW.

Closed application obfuscation, port evasion attack paths

Critical Infrastructure | Southeast Asia

Incorrectly configured vendor capabilities on DMZ rules meant that not all applications on the DMZ were blocked as intended, making the customer susceptible to application obfuscation and port hopping evasion attacks.

Assured SMTP traffic was compliant with NIST 800-53

Finance | East Asia

A portion of the SMTP traffic from trusted to untrusted zones had no threat inspection profiles enabled, despite the NGFW's built-in capability that blocks SMTP-based attacks by default. This exposed the customer to phishing attacks, malicious email attachments and credential harvesting.



Mitigated unauthorized use of service tokens on supply chain repository

Network Security | North America

A supply chain vendor on one of North America's multi-billion dollar market cap companies had inadvertently exposed multiple JWT service account privileged tokens on the public Internet. This exposed the company to unauthorized API access, and catastrophic multi-tenant breach.